Fiscal Year:	FY 2005	**Task Last Updated:**	FY 10/07/2004
PI Name:	Dantzig, Jonathan A Ph.D.		
Project Title:	Phase Field Modeling of Microstructure Development in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Basics of solidification		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Urbana	**State:**	IL
Zip Code:	61801	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	11/30/2000	**End Date:**	11/30/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Goldenfeld, N. D. (University of Illinois at Urbana-Champaign)		
Grant/Contract No.:	NAG8-1657		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Advanced computational techniques for adaptive grid refinement are used to study microstructural evolution in dendritic solidification, for both pure isolated dendrites and directionally solidified alloys. These methods are applied to the study of experiments performed by other PI's in the NASA program to improve the understanding and interpretation of the phenomena. We have two target collaborations: the groups studying dendritic growth in pure materials, including the Isothermal Dendritic Growth Experiment (IDGE), and the Transient Dendritic Growth Experiment (TDSE); and groups studying alloy solidification. By working with these researchers we can broaden the impact of our research findings. The objective of this task is to allow us to facilitate collaboration between our research group and those of other NASA investigators working on dendritic growth. We apply advanced computational techniques for adaptive grid refinement to study microstructural evolution in dendritic solidification, for both pure isolated dendrites and directionally solidified alloys. This work has enabled us to compute dendritic microstructures using both realistic material parameters and		

	experimentally relevant processing conditions, thus allowing for the first time direct comparison of phase field computations with laboratory observations. Many of the NASA-sponsored experimental programs seek to understand the mechanisms for pattern formation during dendritic growth. We take a modeling approach to understanding these same phenomena. By working together with experimentalists, we are able to do a better job at modeling, and we can improve their ability to interpret their experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	This year, our efforts focused on the study of directional solidification in confined spaces, such as between microscope slides, where many experiments are performed. A wide range of conditions was examined, and some important results were found. We determined that when the slide spacing becomes on the same order as the diffusion field near the tips of the advancing front, there is a strong effect on the evolution of the microstructure. In particular, we find that the structure transforms from 3-D dendrites to 2-D dendrites, and that there is a discontinuous change in the pattern selection, as evidenced by a sudden change in dendrite tip radius.
Bibliography Type:	Description: (Last Updated: 04/03/2014)
Books/Book Chapters	Dantzig, J.A. "Solidification processes: From dendrites to design" in "Continuum scale simulation of Engineering Materials: Fundamentals-Microstructures-Process applications" Ed. Raabe, D.; Roters, F.; Chen, L.-Q. Wiley, in press , Oct-2004
Books/Book Chapters	Athreya, B.P.; Dantzig, J.A. "Dendritic growth in confined spaces" in "Solidification processes and microstructures: A symposium in honor of Wilfried Kurz" Ed. Rappaz, M.; Beckermann, C.; Trivedi, R. TMS-AIME, 357-368 , Feb-2004